Differential Equation Descriptions System Characteristics

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Key Concepts

1) The solution to a differential equation gives insight into the relationship between the difference equation parameters and the corresponding system characteristics.

2) Difference equation

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

3) Output $y(t) = y_s(t) + y_t(t)$

a) Steady-state response $y_s(t)$. If $x(t) = e^{st}$ then $y_s(t) = H(s)e^{st}$ where

$$H(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$$

b) Transient response $y_t(t) = \sum_{\ell=1}^{N} c_\ell e^{\tilde{s}_\ell t}$ where $\tilde{s}_\ell$ are the N (distinct) roots of the characteristic equation

$$\sum_{k=0}^{N} a_k s^k = 0$$

and c_ℓ are chosen so $y(t)$ satisfies initial conditions $y(0^-), \frac{d}{dt}y(t)\Big|_{t=0^-}, \ldots, \frac{d^{N-1}}{dt^{N-1}}y(t)\Big|_{t=0^-}$.

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4) Stable systems have $\text{Re}\{\tilde{s}_\ell\} < 0$ for all ℓ. If $\text{Re}\{\tilde{s}_\ell\} > 0$, then $|y(t)|$ can grow without bound due to $|e^{\tilde{s}_\ell t}| \to \infty$ as $t \to \infty$.

5) Under steady-state conditions the output of the system is the same form as the input. For example, sinusoidal input produces a sinusoidal output of the same frequency.

6) The transient response describes how the system transitions from its initial state (at rest or having stored energy) to steady-state behavior. It depends on $\tilde{s}_\ell$, the roots of the characteristic equation.

    a) If $\tilde{s}_\ell$ is real, then it contributes a real exponential to the output.

i) If $\text{Re}\{\tilde{s}_\ell\} < 0$, the exponential decays as t increases.

ii) The response decays slowly for $\text{Re}\{\tilde{s}_\ell\} \approx 0$ and decays quickly for $\text{Re}\{\tilde{s}_\ell\} << 0$.

b) If $\tilde{s}_\ell$ is complex, it contributes exponentially damped sinusoidal terms to the output.

i) The rate of exponential decay depends on $\text{Re}\{\tilde{s}_\ell\}$ with slow decay for $\text{Re}\{\tilde{s}_\ell\} \approx 0$ and fast decay for $\text{Re}\{\tilde{s}_\ell\} << 0$.

ii) The imaginary part of $\tilde{s}_\ell$ determines the frequency of the exponentially damped sinusoid.

iii) If the coefficients of the difference equation are real valued, then the $\tilde{s}_\ell$ occur in complex conjugate pairs.

c) The real part of the roots of the characteristic equation ($\text{Re}\{\tilde{s}_\ell\}$) determine the response time of the system. Response time decreases as $\text{Re}\{\tilde{s}_\ell\}$ decreases. Systems with $\text{Re}\{\tilde{s}_\ell\} \approx 0$ for any ℓ generally have slow response times.

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Video Screenshots

The parameters of the differential equation tell us a lot about the behavior of the system.

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

Initial conditions $y(t)\big|_{t=0^-}$, $\frac{d}{dt}y(t)\big|_{t=0^-}$, $\frac{d^2}{dt^2}y(t)\big|_{t=0^-}$ ... $\frac{d^{N-1}}{dt^{N-1}}y(t)\big|_{t=0^-}$

Solution: $y(t) = \overset{(1)}{y_s(t)} + \overset{(2)}{y_t(t)}$

1) Steady state response: as if input is applied forever

2) Transient response: transition from initial conditions to steady state behavior

Steady-state response:

$$x(t) = e^{st} \longrightarrow y_s(t) = H(s)e^{st}; \qquad H(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$$

By linearity- $x(t) = \alpha_1 e^{s_1 t} + \alpha_2 e^{s_2 t} \longrightarrow y_s(t) = \alpha_1 H(s_1) e^{s_1 t} + \alpha_2 H(s_2) e^{s_2 t}$

Special cases:

1) Constant input: $(s=0)$ $x(t) = \alpha \longrightarrow y_s(t) = \alpha H(0)$

2) Sinusoidal input: $(s_1 = j\Omega, s_2 = -j\Omega)$ $x(t) = \frac{\alpha}{2} e^{j\Omega t} + \frac{\alpha}{2} e^{-j\Omega t}$

$$y_s(t) = \frac{\alpha}{2} H(j\Omega) e^{j\Omega t} + \frac{\alpha}{2} H(-j\Omega) e^{-j\Omega t} \qquad \text{if } \{a_k, b_k\} \text{ real, } H^*(j\Omega) = H(-j\Omega)$$

$$\Rightarrow y_s(t) = A\cos(\Omega t) + B\sin(\Omega t)$$

3) Real exponential: $(s = \beta, \text{real})$ $x(t) = \alpha e^{\beta t} \longrightarrow y(t) = \alpha H(\beta) e^{\beta t}$

- Output is the same form as the input
- System may discriminate inputs (amplify some, attenuate others)

Transient response:

$$y_t(t) = \sum_{\ell=1}^{N} C_\ell e^{\tilde{s}_\ell t}, \quad \tilde{s}_\ell \text{ are roots of } a_N s^N + a_{N-1} s^{N-1} + \ldots + a_1 s + a_0 = 0$$

Repeated roots: $\tilde{s}_i$ repeated d times $\rightarrow e^{\tilde{s}_i t}, t e^{\tilde{s}_i t}, t^2 e^{\tilde{s}_i t}, \ldots t^{d-1} e^{\tilde{s}_i t}$

Transient response characteristics are governed by roots of char. eqn!

What does $e^{\tilde{s}_\ell t}$ look like?

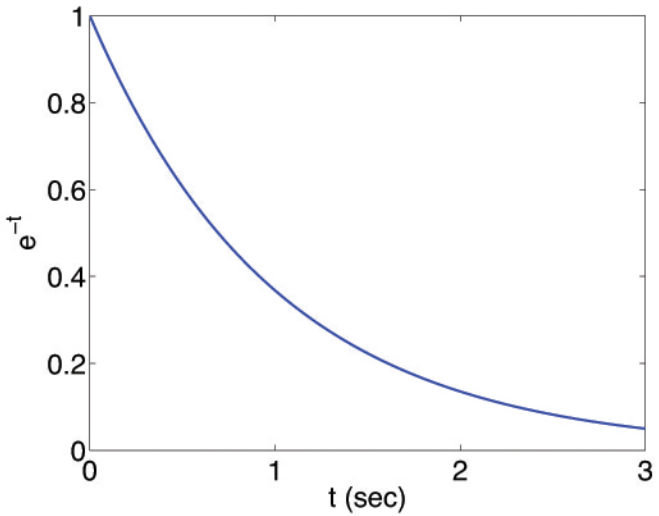
$$e^{\tilde{s}_\ell t} = e^{\mathrm{Re}\{\tilde{s}_\ell\} t} e^{j \mathrm{Im}\{\tilde{s}_\ell\} t}$$

1) $\tilde{s}_\ell$ real $\Rightarrow$ real exponential
2) $\tilde{s}_\ell$ imaginary $\Rightarrow$ complex sinusoid
3) $\tilde{s}_\ell$ complex $\Rightarrow$ exponentially damped sinusoid

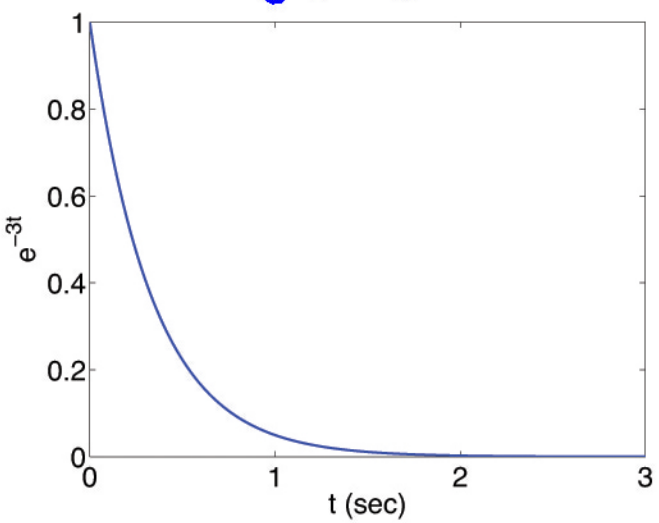
$\mathrm{Re}\{\tilde{s}_\ell\} > 0 \Rightarrow$ unstable system

Example Waveforms: $e^{\tilde{s}t}$

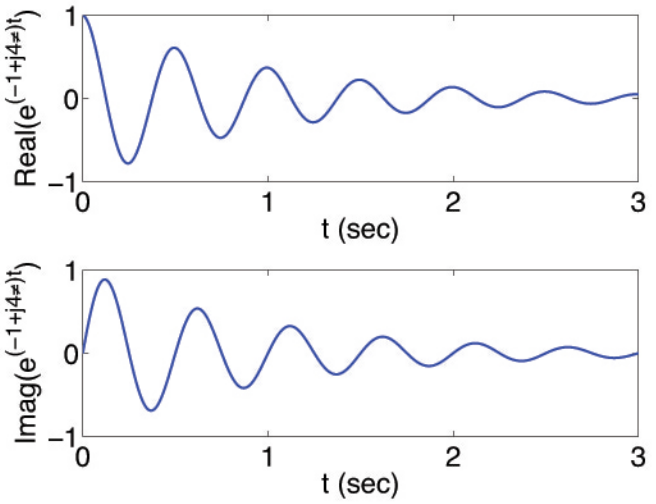
$\tilde{s} = -1$



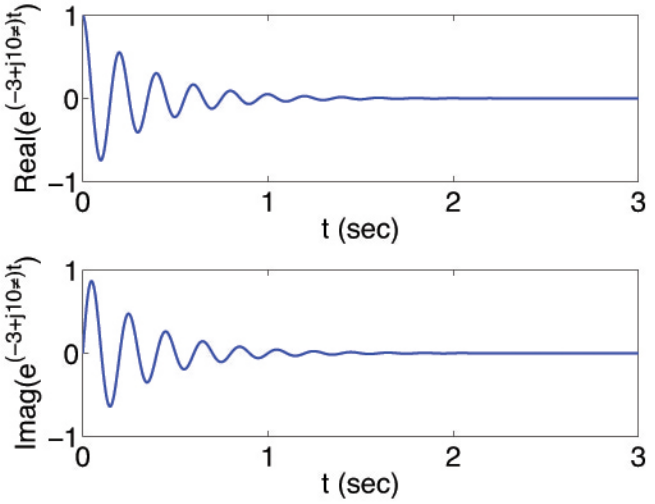
$\tilde{s} = -3$



$\tilde{s} = -1 + j4\pi$



$\tilde{s} = -3 + j10\pi$



Faster response as $\mathrm{Re}\{\tilde{s}\}$ decreases

Example: $\frac{d^2}{dt^2}y(t) + a_1\frac{d}{dt}y(t) + a_0 y(t) = b_0 x(t)$; $x(t) = u(t)$

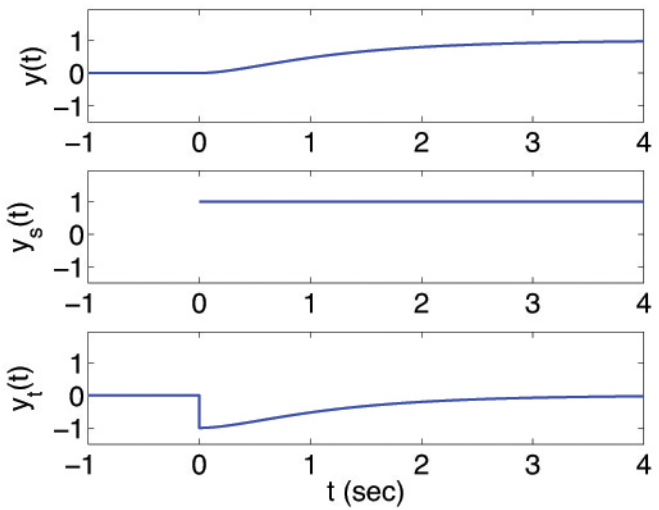
$y(0^-) = 0$, $\frac{d}{dt}y(t)\Big|_{t=0^-} = 0$

$$H(s) = \frac{b_0}{s^2 + a_1 s + a_0} \quad ; \quad y_s(t) = H(0) = \frac{b_0}{a_0}$$

$$y(t) = \frac{b_0}{a_0} + c_1 e^{\tilde{s}_1 t} + c_2 e^{\tilde{s}_2 t}, \quad t > 0$$

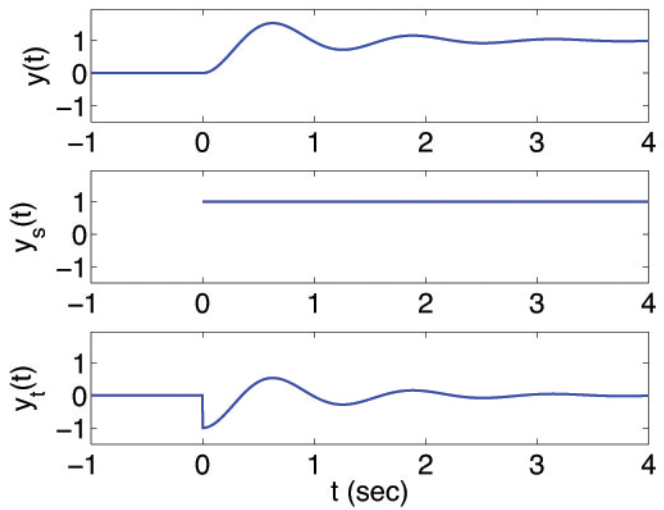
$a_1 = 4,\ a_0 = 3,\ b_0 = 3$

$\tilde{s}_1 = -1$, $\tilde{s}_2 = -3$

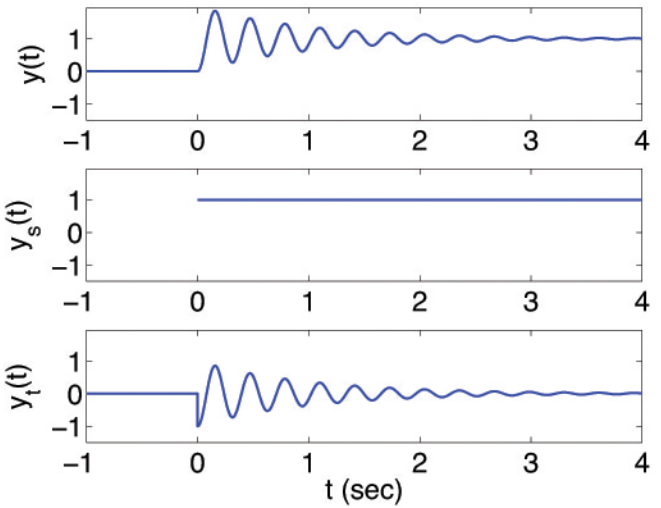


$a_1 = 2,\ a_0 = 26,\ b_0 = 26$

$\tilde{s}_1 = -1 + j5$, $\tilde{s}_2 = -1 - j5$



$\tilde{s}_{1,2} = -1 \pm j20$



$\tilde{s}_{1,2} = -3 \pm j20$

